

A Multimodal Approach to Fake News Detection: Integrating BERT and ResNet50

Abstract -- *The increasing rate of which misinformation is spread across diverse domains poses a significant threat to public health and security. This study proposes a multimodal approach to fake news detection, effectively integrating the strengths of advanced language understanding and visual feature extraction. Recognizing that fake news are often both deceptive textual narratives and misleading imagery, our methodology combines textual information processed by a pre-trained BERT model with visual cues extracted by a ResNet50 architecture. Utilizing a Fakeddit dataset that includes both textual and visual modalities, we develop and evaluate this integrated model. Our evaluation, encompassing accuracy, precision, recall, and F1-score, aims to demonstrate how the fusion of these distinct modalities enhances detection capabilities. This integration is anticipated to yield improved generalization and robustness compared to other approaches, highlighting the synergistic potential of combining deep learning models for comprehensive and adaptive fake news detection across complex, real-world scenarios.*

Keywords—ResNet50, BERT, Multimodal, Fakeddit Dataset

I. INTRODUCTION

In recent years, the rapid spread of misinformation across various domains, including health and politics, has caused major challenges to news integrity. A study from MIT[1] revealed that false news are 70% more likely to be retweeted than the actual true stories, which travels at six times faster to reach 1500 people. This facilitates the widespread of inaccurate information leading to challenging containment, such as the case during COVID-19 pandemic where 46,000 daily twitter contained misinformation. This leads to confusion to people who happened to read the tweets and having knowledge of a false health information. This eroded the public confidence, and therefore made the containment of the crisis more challenging. With this information in mind, it demonstrates how a single misinformation can be passed onto many individuals and might lead endangered lives.

Researching fake news detection is significant as it plays a very crucial role in restoring public trust in media and institutions. This matter has a huge impact on society, as fake news can influence public opinion, sway elections, leading to a public disturbance, making effective detection mechanisms essential for ensuring that accurate information reaches the public, especially during critical moments. Additionally, there are plenty of advancements and innovations in detection algorithms contributing to the fields of machine learning and natural language processing.

The ability of accurately detecting misinformation has become more critical. While deep learning models like ResNet50 have demonstrated remarkable performance in image tasks, accurately detecting misinformation often requires understanding both textual content and image content, therefore a multimodal approach would be more suitable. This paper proposes a method that combines the

strengths of BERT for sophisticated text understanding and ResNet50 for robust visual feature extraction, aiming to enhance the accuracy and robustness of fake news detection across complex, real-world scenarios.

The increasing spread of misinformation will further impact society and so is the significance of developing a reliable and accurate automated detection of fake news through different modal such as in a text-based tasks and image-based tasks. Fake news often exploits both domain-specific language and deceptive imagery to mislead audiences. Therefore, a robust and accurate detection system necessitates a multimodal approach capable of analyzing both textually and visually. The integration of advanced models like BERT for understanding textual tasks and ResNet50 for extracting features from associated images is crucial. By combining these powerful architectures, our approach aims to create a flexible and adaptable system that can generalize effectively across diverse domains. This enhances the reliability of automated misinformation detection without sole reliance on large labeled datasets for each new domain.

Current research on misinformation detection has explored various machine learning and deep learning techniques, including transformer-based models (e.g., BERT and RoBERTa). Others have proposed multi-modal detection techniques that combine text and image analysis to enhance accuracy. However, while ResNet50s have been widely applied in text classification tasks, their effectiveness in domain-specific misinformation detection can still be further explored.

Our models are trained and evaluated on a dataset comprising both textual and visual content. Evaluation will utilize standard metrics including accuracy, precision, recall, F1-score, and statistical analysis to assess the significance of the combined model's performance. This research contributes to a better understanding of the synergistic role of integrating transformer-based models like BERT with convolutional neural networks such as ResNet50 for enhanced, robust multimodal fake news detection across diverse real-world scenarios. We compared the results with other models such as using SVM, Multinomial Naïve Bayes, Logistic Regression, and Random Forest Classifier.

II. RELATED WORKS

The rapid spread of fake news has become a critical challenge in the digital age, posing significant advancements in detection methodologies. Researchers have increasingly turned to deep learning techniques to enhance the accuracy and effectiveness of these detection systems. While there are research that are focused on establishing basic frameworks for identifying misinformation, the complexity of fake news now

requires more sophisticated approaches capable of adapting to a variety of contexts and content types.

In recent years, the growing complexity of fake news, often having both textual and visual elements, has driven significant research into multimodal detection methodologies. There are various studies that focused on advanced techniques to effectively integrate information from diverse sources. Among these, Segura-Bedmar and Alonso-Bartolome [2] provided a comprehensive overview of multimodal fake news detection, highlighting the critical aspects and emerging trends in the field. Furthermore, Xue et al. [3] explored approaches to detecting fake news by examining the consistency across multimodal data, highlighting how dependencies between different modalities can serve as crucial indicators of misinformation. Building on this, Wang et al. [4] introduced Fine-Grained Multimodal Fusion Networks (FMFN), highlighting the importance of detailed and correct implementation of integration strategies to enhance detection accuracy.

Further advancing the field, Jing et al. [5] proposed multimodal fake news detection via progressive fusion networks, which progressively combine features from different modalities, allowing for a more dynamic and adaptive understanding of deceptive content. Collectively, these works underscore the critical shift from unimodal to multimodal detection, demonstrating how advanced fusion mechanisms and the exploration of inter-modal consistency are pivotal in developing more robust and effective fake news detection systems capable of addressing the multi-nature of online misinformation containing image and text representation.

Research by Zhang et al. [6] demonstrates that using auxiliary mechanisms allows models to better capture the multi-sided nature of misinformation. This enhances their ability to identify misleading content across various domains. This integration of different data types not only enriches the model's understanding but also helps in recognizing the diverse ways through which misinformation can be presented. In this context, Ying et al. [7] introduced a *Multi-Level Multi-Modal Cross-Attention Network* that incorporates multimodal cues at various attention levels, allowing for a more deeper understanding of how misinformation are spread through both text and images.

As the field evolves, addressing multi-domain challenges has become increasingly important. Innovative techniques that employ soft-labeling, as discussed by Wang et al. [8], enable models to express varying degrees of confidence in their predictions—particularly vital in political or emotionally charged contexts where distinctions between fact and opinion are often blurred. This adaptability is crucial for navigating the challenging misinformation detection. Furthermore, Silva et al. [9] underscore the necessity of tailoring models to the unique characteristics of different sectors, such as health and finance. Optimizing a model for one domain may not yield effective results in another, this encourages the need for focused training and domain-specific fine-tuning. The connection between soft-labeling and domain-aware training provides a more refined and signified response to fake news.

An innovative direction involves the use of prompt-based mechanisms, as explored by Alghamdi et al. [10], which guide models to detect misleading information across different contexts. This method increases the adaptability of detection

systems, allowing them to respond to evolving misinformation strategies in real-time. Similarly, domain- and instance-level transfer learning, as discussed by Wang et al. [11], allows models to adapt to specific events or rapidly changing information landscapes. This also implies to those encountered during political campaigns or public health emergencies.

The integration of multimodal data continues to play a transformative role in fake news detection. Research by Ferdush et al. [12] shows that synthesizing information from various sources significantly improves model performance by offering a broader contextual understanding. This data synthesis helps not only in pinpointing misinformation but also in decoding the larger narratives through which such content spreads. Building on this, Agrawal et al. [13] proposed a hybrid model combining hyperparameter-tuned BERT and ResNet110, improving detection accuracy by effectively analyzing both textual and visual content. Similarly, Almarashy et al. [14] enhanced detection systems through deep neural networks that encode fused multimodal features, enabling richer and more robust misinformation classification.

In the field of fake news detection, deep learning approaches have demonstrated significant performance. For instance, Nan et al. [15] introduced MDFEND, a framework specifically designed for multi-domain fake news detection, addressing the challenge of misinformation spreading across varied contexts. Their work highlights the importance of adaptability in detection systems when faced with diverse textual characteristics. Complementing this, Sharma et al. [16] explored a deep learning approach for the automatic detection of fake news, often encouraging complex embedding techniques like FastText to capture semantic representations of textual content. These studies contribute to the foundational understanding of applying advanced computational models to identify false information. This sets a start for further exploration into more complex, integrated architectures.

Finally, the application of deep learning architectures to automate the detection process, as highlighted by Mridha et al. [17], showcases the potential of neural networks in analyzing complex textual patterns. These have led to noticeable improvements over traditional methods, demonstrating the value of ongoing innovation in this genre. By using advanced techniques and acknowledging the nature of misinformation, researchers are developing a more accurate, adaptable, and scalable detection systems. Furthermore, researchers have increasingly used deep learning techniques to enhance the accuracy and interpretability of these systems. Sap et al. [18] proposed interpretable neural models to explain why certain messages might be identified as fake, highlighting the importance of explainability in modern fake news detection systems. Fake news today is far more complicated and needs innovative methods that can understand different ways people use language. On the other hand, research on adversarial robustness highlights the need for detection models resilient to manipulation attempts. Research by Jin et al. [19] shows that BERT are highly vulnerable to adversarial attacks in natural language processing task. These considerations highlight the challenges that needs to be faced in detection misinformation, motivating ongoing innovation in adaptable and reliable model architectures.

To sum up, the fight against fake news has driven the evolution of detection methodologies that can address the complexities of misinformation across multiple contexts.

Advances in deep learning—particularly through multilingual, multimodal, and multi-level architectures—have significantly improved both the precision and resilience of these systems. Through soft-labeling, domain-specific fine-tuning, auxiliary mechanisms, and cross-attention networks, models are becoming increasingly adept at navigating an ever-evolving misinformation landscape. These developments not only bolster detection but also promote a more informed public and healthier information ecosystem. As fake news tactics grow more sophisticated, so too must our technological and research efforts in safeguarding the integrity of information.

In this study, our multimodal approach utilizes ResNet50, a deep convolutional neural network used for its exceptional performance in visual tasks. Employing residual learning, ResNet50 efficiently processes complex image data to extract salient visual features. Within our framework, ResNet50's role is specifically to analyze the image component of news items, identifying visual cues that may contribute to or contradict the authenticity of the associated textual information. This robust visual analysis by ResNet50 complements the deep textual understanding provided by BERT, forming a comprehensive system for fake news detection.

III. METHODOLOGY

Our methodology adopts a quantitative approach to compare the performance of a combined ResNet50 and BERT model on the detection accuracy of misinformation in domain-specific contexts, specifically in health and politics.

A. Data Collection

1) Dataset

Our dataset for this study will be sourced from Fakeddit, a large-scale multimodal fake news dataset designed to support comprehensive research in this field. Fakeddit [20] has over 1 million samples derived from Reddit posts. This dataset is particularly valuable due to its rich multimodal nature, encompassing not only textual content but also associated image data, extensive metadata, and user comments. Each sample is labeled across various truthfulness categories, supporting 2-way, 3-way, and 6-way classification, which allows for deep and thorough analysis of misinformation. Its comprehensive structure and significant size make Fakeddit an ideal resource for developing and evaluating advanced multimodal fake news detection models.

2) Data Preprocessing

The data preprocessing phase is crucial for enhancing the accuracy of misinformation detection. It begins with handling potential missing values by ensuring that text fields are consistently treated as strings. Next, text cleaning is performed, which involves converting all text to lowercase, removing punctuation, numerical digits, and extraneous spaces. This process helps standardize the textual data for consistent model training. Finally, any records where the text becomes empty after this cleaning are removed to ensure valid inputs and prevent inaccuracies during model training.

As for the image, it involves a series of transformations to ensure images are consistent and optimized for the network. The first step is to resize all the images to a uniform dimension, such as 224x224 pixels, which is necessary because models expect a fixed input size. Pixels are then

scaled to a common range, like 0.0-1.0, which aids in stable numerical processing. Finally, normalization adjusts pixel values across color channels using pre-defined means and standard deviations, ensuring the image data distribution aligns with what pre-trained models expect. This comprehensive preprocessing pipeline is vital for the model to effectively learn from diverse visual inputs and achieve reliable performance.

B. Proposed Method

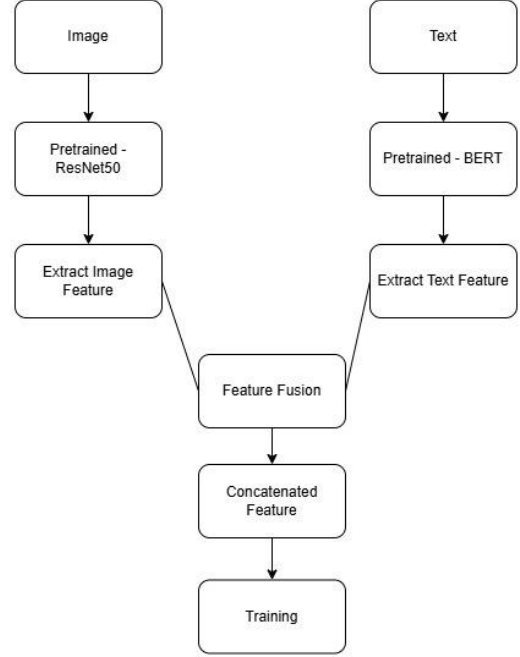


Fig 1. Method Workflow

In this research, we will be combining a ResNet50 model and BERT model, *Figure 1* depicts the proposed method for the multimodal fake news detection. The proposed method involves feature extraction, combined model training and optimization, and evaluation. For the textual data, Feature extraction utilizes BERT to convert textual data into numerical vector representations. For visual data, the ResNet50 model will extract powerful image features. These distinct textual and visual features will then be combined to form a comprehensive multimodal representation.

During the training and optimization, we will use binary cross-entropy as the loss function and also implement the Adam optimizer to minimize the loss function. Binary cross-entropy serves as a scoring rule. It measures how wrong a model's prediction is, heavily penalizing confident incorrect guesses and thereby guiding the model during training to make its predictions as accurate as possible.

We used PyTorch for this experiment as it provides the necessary tools and flexibility to construct a complex neural network architecture. This allows the seamless integration of BERT and ResNet50. Furthermore, PyTorch enables the loading, preprocessing and batching of multimodal dataset. This works well with our chosen dataset of Fakeddit. Given that the Fakeddit dataset will be used, it will be appropriately partitioned into training, validation, and testing sets to ensure robust model development and unbiased evaluation. The

Fakeddit dataset are already split into those mentioned. This integrated setup allows us to effectively explore and demonstrate the enhanced capabilities of a multimodal system in detecting misinformation.

C. Evaluation

Several performance metrics were employed to validate the Fake News Detection model, including accuracy, precision, recall and F1-score. These metrics provide a comprehensive evaluation of the model's ability to distinguish between real and fake news. To optimize the model, the model was trained using the Adam optimizer with a learning rate of 2×10^{-5} . To prevent overfitting and enhance generalization, dropout layers were strategically applied: a 0.2 dropout rate was used in both the BERT and ResNet50 streams, and a 0.3 dropout rate was applied within the final classification head. Additionally, the layers of the pre-trained BERT and ResNet50 models were initially frozen, serving as a form of regularization. The training process was conducted over 5 epochs with a batch size of 48, and early stopping was implemented with a patience of 3 epochs based on validation loss.

IV. RESULTS AND DISCUSSION

The Integrated Multimodal Model (ResNet50 + BERT) consistently demonstrates strong performance against traditional models. It is able to Achieve an Accuracy of 82.834% and an F1-Score of 80.151%, its effectiveness in fake news detection is evident. Notably, the model's Recall stands out as remarkably high at 87.386%, significantly surpassing the performance of all traditional models evaluated. This exceptional recall is of paramount importance in the domain of fake news detection, as it signifies the model's superior ability to correctly identify genuine instances of misinformation, thereby minimizing critical false negatives—a key objective in mitigating the societal impact of false narratives. While the model's Precision at 74.020% is observed to be lower than some traditional counterparts, the F1-Score of 80.151% indicates an overall strong and balanced performance.

This balance is crucial, as the F1-Score combines precision and recall, providing a more accurate measure of the model's effectiveness in real-world scenarios where both correctly identifying fake news and avoiding misclassifying true news are vital. This superior performance is a direct and undeniable consequence of the combined model's unique capacity to effectively integrate complementary insights derived from both textual narratives (leveraging BERT's profound linguistic understanding) and associated visual elements (through ResNet50's robust image analysis).

TABLE I. EXPERIMENTAL RESULTS USING TF-IDF FOR THE OTHER MODELS.

Model	Accuracy	Precision	Recall	F1-Score
Combined Model (ResNet50 + BERT)	82.834%	74.020%	87.386%	80.151%
Multinomial Naïve Bayes	79.912%	79.750%	79.735%	79.912%
Logistic Regression	82.775%	82.668%	82.635%	82.775%
Support Vector Machine	82.744%	82.639%	82.647%	82.744%
Random Forest	82.151%	82.054%	82.077%	82.151%

TABLE II. EXPERIMENTAL RESULTS USING COUNTVECTORIZER FOR THE OTHER MODELS

Model	Accuracy	Precision	Recall	F1-Score
Combined Model (ResNet50 + BERT)	82.834%	74.020%	87.386%	80.151%
Multinomial Naïve Bayes	80.440%	80.562%	80.489%	80.440%
Logistic Regression	82.969%	82.956%	82.714%	82.969%
Support Vector Machine	83.126%	83.083%	82.904%	83.126%
Random Forest	81.070%	80.949%	80.841%	81.070%

Table I gives us a clear look at how our traditional models perform when we use TF-IDF vectorization. It also lets us compare these results with the performance of our new, combined model. Then, Table II shows us the results for the same traditional models, but this time they are using CountVectorizer to turn the text into numbers.

When we look closely at the numbers, we can see something interesting. For models like Naive Bayes, Logistic Regression, and SVM, their accuracy actually went up a little bit when we switched from TF-IDF to CountVectorizer. However, this increase in accuracy was really, really small. It was such a tiny change that, for these specific traditional models, it doesn't make a big difference which way you choose to vectorize the text. Both methods give very similar results, so you don't need to worry too much about picking one over the other for these models.

The result here shows that the accuracy obtained is largely similar to that observed in many other multi-modal models presented by various researchers in the field. While the overall performance aligns closely with the current researches, there are indeed a few outliers, such as a specific research that is conducted by Zhang et al. [6], which demonstrated a noticeably higher accuracy, resulting in a figure of around 90% in their multi-modal fake news detection framework. This minor divergence from such top-tier outliers, where our model shows a generally comparable but sometimes slightly less optimized accuracy, may be partly attributed to the

intrinsic nature of our BERT model implementation. Specifically, our approach utilizes a base pre-trained BERT model. While highly effective for general language understanding, this standard pre-trained configuration might not fully capture the complex deceptive patterns present across diverse textual and visual information in fake news as effectively as those highly optimized outlier systems, which may employ more complex BERT variants or fusion strategies.

V. CONCLUSION

In conclusion, this project successfully developed and evaluated a multimodal fake news detection system that integrates the complementary strengths of the BERT model for textual analysis and the ResNet50 model for visual feature extraction. Our experiments demonstrated that this combined approach had a superior performance in detecting misinformation compared to unimodal baselines that rely solely on text or image. The findings underscore the critical importance of having both textual content and imagery content, as the fusion of these modalities provides a more robust and comprehensive understanding of complex misinformation. This research contributes valuable insights into the efficiency of integrated deep learning architectures for tackling the pervasive challenge of fake news.

Future work could investigate smarter ways to combine information from text and images. We could explore methods that let the model understand more deeply how text and images work together. It is important to test this multimodal system on bigger and more varied datasets. This means using different kinds of fake news from more sources and in various languages, which will help the model work better and be more useful in real-life situations. Lastly, we could try using newer and more advanced text models than the basic BERT, like RoBERTa or XLNet. These stronger models can understand text even better, which would help make our fake news detection even more accurate and efficient as fake news keeps changing.

REFERENCES

- [1] Study: On Twitter, false news travels faster than true stories. (2018, March 8). MIT News | Massachusetts Institute of Technology. <https://news.mit.edu/2018/study-twitter-false-news-travels-faster-true-stories-0308>
- [2] Segura-Bedmar, I., & Alonso-Bartolome, S. (2022). Multimodal Fake News Detection. *Information* (Switzerland), 13(6). <https://doi.org/10.3390/info13060284>
- [3] Xue, J., Wang, Y., Tian, Y., Li, Y., Shi, L., & Wei, L. (2021). Detecting fake news by exploring the consistency of multimodal data. *Information Processing and Management*, 58(5). <https://doi.org/10.1016/j.ipm.2021.102610>
- [4] Wang, J., Mao, H., & Li, H. (2022). FMFN: Fine-Grained Multimodal Fusion Networks for Fake News Detection. *Applied Sciences* (Switzerland), 12(3). <https://doi.org/10.3390/app12031093>
- [5] Jing, J., Wu, H., Sun, J., Fang, X., & Zhang, H. (2023). Multimodal fake news detection via progressive fusion networks. *Information Processing and Management*, 60(1). <https://doi.org/10.1016/j.ipm.2022.103120>
- [6] Zhang, D., Shan, G., Lee, M., Zhou, L., & Fu, Z. (2025). MT-GPD: A Multimodal Deep Transfer Learning Model Enhanced by Auxiliary Mechanisms for Cross-Domain Online Fake News Detection. *Production and Operations Management*. <https://doi.org/10.1177/10591478251319686>
- [7] Ying, L., Yu, H., Wang, J., Ji, Y., & Qian, S. (2021). Multi-Level Multi-Modal Cross-Attention Network for Fake News Detection. *IEEE Access*, 9, 132363–132373. <https://doi.org/10.1109/ACCESS.2021.3114093>
- [8] Wang, D., Zhang, W., Wu, W., & Guo, X. (2023). Soft-Label for Multi-Domain Fake News Detection. *IEEE Access*, 11, 98596–98606. <https://doi.org/10.1109/ACCESS.2023.3313602>
- [9] Silva, A., Luo, L., Karunasekera, S., & Leckie, C. (2021a). Embracing Domain Differences in Fake News: Cross-domain Fake News Detection using Multi-modal Data. <https://ojs.aaai.org/index.php/AAAI/article/view/16134>
- [10] Alghamdi, J., Lin, Y., & Luo, S. (2024). Cross-Domain Fake News Detection Using a Prompt-Based Approach. *Future Internet*, 16(8). <https://doi.org/10.3390/fi16080286>
- [11] Wang, D., Zhu, Y., Li, J., Nan, Q., Sheng, Q., Shi, Y., & Cao, J. (n.d.). Improving Fake News Detection of Influential Domain via Domain- and Instance-Level Transfer. <https://doi.org/10.48550/arXiv.2209.08902>
- [12] Ferdush, J., Kamruzzaman, J., Karmakar, G., Gondal, I., & Das, R. (2025). Cross-Domain Fake News Detection Through Fusion of Evidence from Multiple Social Media Platforms. *Future Internet*, 17(2). <https://doi.org/10.3390/fi17020061>
- [13] Agrawal, C., Pandey, A., & Goyal, S. (2024). Multimodal fake news detection using hyperparameter-tuned BERT and ResNet110. *International Journal of Advanced Technology and Engineering Exploration*, 11(114), 759–772. <https://doi.org/10.19101/IJATEE.2023.10102314>
- [14] Hashim Jawad Almarashy, A., Feizi-Derakhshi, M. R., & Salehpour, P. (2024). Elevating Fake News Detection Through Deep Neural Networks, Encoding Fused Multi-Modal Features. *IEEE Access*, 12, 82146–82155. <https://doi.org/10.1109/ACCESS.2024.3411926>
- [15] Nan, Q., Cao, J., Zhu, Y., Wang, Y., & Li, J. (2021). MDFEND: Multi-domain Fake News Detection. *International Conference on Information and Knowledge Management, Proceedings*, 3343–3347. <https://doi.org/10.1145/3459637.3482139>
- [16] Sharma, D. M., Bhattacharyya, P., Sangal, R., Saikh, T., De, A., Ekbal, A., & Bhattacharyya, P. (2019). A Deep Learning Approach for Automatic Detection of Fake News. *NLPAL*. <https://aclanthology.org/2019.icon-1.27/>
- [17] Mridha, M. F., Keya, A. J., Hamid, M. A., Monowar, M. M., & Rahman, M. S. (2021). A Comprehensive Review on Fake News Detection with Deep Learning. *IEEE Access*, 9, 156151–156170. <https://doi.org/10.1109/ACCESS.2021.3129329>
- [18] Sap, M., Gabriel, S., Qin, L., Jurafsky, D., Smith, N. A., & Choi, Y. (2020). Social Bias Frames: Reasoning about Social and Power Implications of Language. <https://doi.org/10.18653/v1/2020.acl-main.486>
- [19] AAAI. (2023, April 28). Is BERT really robust? A strong baseline for natural language attack on text Classification and Entailment - AAAI. <https://aaai.org/papers/08018-is-bert-really-robust-a-strong-baseline-for-natural-language-attack-on-text-classification-and-entailment/>
- [20] *Papers with Code - r/Fakeddit: A New Multimodal Benchmark Dataset for Fine-grained Fake News Detection. (2019, November 10).* <https://paperswithcode.com/paper/rfakeddit-a-new-multimodal-benchmark-dataset>